BENJAMIN LAPPALAINEN

Creative Technologist & Software Developer: Networked Systems, Interactive Media & Computer Vision
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Engineer-trained technologist with 6+ years building networked and real-time systems, and a recent focus on teaching them. Customer-facing technical experience across partners, clients and workshop cohorts. Combines hands-on software development (TouchDesigner, Python, JavaScript, C++) with product and infrastructure experience from 5 years at a consumer hardware company, including leading IoT protocol certification. Comfortable owning a project end-to-end: scoping, architecture, implementation, and delivery under deadline.

TECHNICAL SKILLS

Networking & Infrastructure: Tailscale, TCP/UDP, IPv4/IPv6, LAN configuration & administration, Cloudflare (Tunnels, DNS, Turnstile), AWS, Thread, Matter protocol, self-hosted Git, NAS/homelab administration
Languages: Python, JavaScript, C++, Lua, GLSL, C, Bash
Real-Time & Creative Coding: TouchDesigner, OSC, MIDI, WebGL, p5.js, shader/GLSL pipelines, Godot
Computer Vision & ML: OpenCV, MediaPipe, ONNX, MoveNet pose estimation, PyTorch, TFLearn
Sensors & Hardware: ArtNet & DMX (lighting/show control), depth & CV cameras (Microsoft Kinect, Orbbec Femto Bolt, Luxonis OAK-D), LIDAR and ultrasonic distance sensors
AI-Assisted Coding: Claude Code, LM Studio, agentic coding workflows, technical review of AI-generated code
Tools & Platforms: Git, Jira, Confluence, Bitbucket, Adobe Creative Suite, Obsidian

EXPERIENCE

Creative Technologist & Software Developer — Independent 2024 – Present

- Provided technical support for a client installation at Toronto Metropolitan University, reverse-engineering an undocumented lighting controller to deliver a working DMX and ArtNet integration.
- Contract software development on AAASeed, a real-time 2D/3D engine for interactive installations, in C++, Lua, and GLSL — architected HTTP networking (via the open-source cpr library) for web API communication, plus other foundational architecture and feature work en route to open-source release.
- Co-creator and technical lead of Peripherous, an open-source machine perception toolkit — directed architecture and AI-assisted implementation (via Claude Code) of its runtime, local message bus, lifecycle supervisor, and OSC output, using MoveNet Multipose (ONNX) for real-time pose estimation.
- Built and distributed TouchDesigner components including a SPAN super-resolution upscaler and macOS-optimized real-time diffusion image generator.
- Research in real-time 3D scene reconstruction pipelines (depth estimation, shared-memory integration) for TouchDesigner, and diffusion-based real-time image pipelines across macOS and Windows.

XR Development Lead — UKAI Projects 2025 – Present

- Develop and lead Digital Bridges, a peer-to-peer networking program teaching researchers and artists to build collaborative systems using Tailscale and OSC.
- Software development and AI system behaviour direction for AI-driven interactive game systems.
- Designed and delivered technical workshops on machine perception and computer vision (MediaPipe) for artist and technologist audiences, including an international iteration in Hsinchu, Taiwan.

Program Lead — soft_launch 2025 – Present

- Co-design and lead a creative technology education program in partnership with InterAccess; design and teach intensives on TouchDesigner ('Reactive Space') and creative coding ('Internet Canvas').

Product Manager — Nanoleaf 2020 – 2024 (plus 2018–2019 internship)

- Managed technical partnerships and software integrations with Apple (Thread & Matter), Google (Matter), Amazon (Matter), Samsung (Matter), Razer (Razer Chroma), and Corsair (iCUE).
- Led certification and technical integration for the Matter smart-home protocol, successfully certifying the first Matter-compliant smart lights; developed working expertise in Thread mesh networking.
- Creator and lead developer of Nanoleaf Screen Mirror, a real-time screen-to-light rendering feature shipped across the product line and central to the flagship Nanoleaf 4D hardware product.
- Built the Nanoleaf Orchestrator music visualizer; developed cross-functional technical documentation for engineering, marketing, and support teams.

EDUCATION

BASc, Engineering Science (Aerospace Specialization) — University of Toronto, 2015 – 2020